



A-site and B-site cation doping engineering: A strategy to enhance strain in Pb(Zr, Ti)O₃-based ceramics

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ABSTRACT

Pb(Zr, Ti)O₃ piezoelectric ceramics with high strain and low hysteresis are essential for ultra-precision driving and positioning in advanced manufacturing and control fields. In this study, the doping engineering via dual cations was utilized to improve the strain characteristics of Pb(Zr, Ti)O₃-based ceramics, and thus the ceramics 0.3PbTiO₃-0.65PbZrO₃-0.05(Bi_{0.3}La_{0.7})(Zn_{0.5}Ti_{0.5})O₃-x wt% (La₂O₃+Nb₂O₅) were probed. Results show that excellent comprehensive performance is fulfilled for the ceramic with x = 5, i.e., a high strain $S = 0.45\%$, low strain hysteresis $H_y = 16\%$, and high piezoelectric coefficient $d_{33} = 585$ pC/N. For modified ceramics, the large electrostrictive strain stems from the synergistic role of lattice distortion and transformation of electric domain morphology due to cations doping. This study demonstrates a productive cation co-doping engineering to enhance the electrostrictive strain in Pb(Zr, Ti)O₃-based ceramics, which enables the ceramics to have good features applied in piezoelectric actuators.

1. Introduction

Piezoelectric materials show wide application prospects in aerospace, medical instruments, and various electronic products [1–5]. Among them, achieving large electrical strain in piezoelectric materials is of great importance for actuator applications [6–8]. PbTiO₃-PbZrO₃ (PT-PZ) piezoelectric ceramic is one of the most promising candidates for micro-displacement systems in precision instruments for its excellent comprehensive performances. However, the low strain (0.1–0.2 %) of commercial piezoelectric materials cannot be applied to the high-precision fields. It is necessary to enhance their electrical strain to achieve broader commercial applications and address the limitations of current lead-based piezoelectric materials.

It is widely acknowledged that the origins of strain in piezoelectric ceramics come from both extrinsic factors, such as domain switching, and intrinsic factors, such as lattice distortion. Currently, there are four common strategies to enhance the strain, including creating morphotropic phase boundaries (MPB), enhancing relaxor properties, establishing local structural heterogeneity, and constructing defect dipoles by ionic doping [9–13]. Doping ions with different radii cause varying degrees of lattice distortion, where a reduction of domain size caused by certain lattice distortion in perovskite crystals enhances the strain.

Therefore, domain reversal becomes easier with an applied electric field [14,15]. The performance of piezoelectric ceramics is affected by the lattice structure and domain wall movement. However, there is a lack of comprehensive studies on their internal and external interactions.

The addition of La³⁺ ions in perovskite piezoelectric ceramics has been demonstrated to have a substantial impact on the piezoelectric strain characteristics. This enhancement makes them suitable for a diverse array of piezoelectric actuator applications [16]. For instance, Zhang et al. [8] obtained a 0.23 % strain by incorporating La³⁺ ions into PIN-PZ-PT ceramics. It is shown that the A-site substitution is beneficial for enhancing heterogeneity and relaxation characteristics. In Tai's study [17], BiFeO₃-BaTiO₃ ceramics were synthesized with La³⁺ ion doping. The research revealed that the incorporation of La³⁺ ions led to an increase in the amounts of domains and facilitated greater displacements of Bi³⁺ ions, resulting in enhanced displacement generation. This promotes the improvement of strain response. Furthermore, B-site doping can also promote the improvement of piezoelectric properties [18,19]. Yang et al. [20] reveal that the introduction of Nb⁵⁺ ions in piezoelectric ceramics mitigates leakage current while simultaneously enhancing the piezoelectric performance. When ions incorporate into the oxygen octahedral sites, the differences in ionic radii and electro-negativity compared to the original ions lead to an asymmetric

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environment. This asymmetry induces a distortion in the normal oxygen octahedral structure, driving it into a condensed electron state. This distortion, in turn, facilitates energy reduction and ultimately stabilizes the material. However, the aforementioned report used methodologies that consider either extrinsic or intrinsic contributions, which may be insufficient to improve the electrical performance. [15]. To achieve a great improvement in electrical properties, it is necessary to consider both types of contributions simultaneously.

The illustration in Fig. 1 demonstrates that the co-doping of La^{3+} and Nb^{5+} ions induces notable lattice distortion. The domain wall moves rapidly under the influence of this distortion and reduces the correlated length of the polar coupling, ultimately leading to a finer *S-E* (strain-electric field) curve. Fig. 1(b) explains the relationship between the high-voltage electrical strain and the microscopic domain structure, showing the schematic diagram of the evolution of long-range ferroelectric domains (Figs. 1(b), (b1-b3)), PNRs with intermediate ferroelectric order are (Figs. 1(b), (b4-b6)), and PNRs (Figs. 1(b), (b7-b9)) for the samples before, during, and after the applied of electric fields, respectively. In addition, the presence of La^{3+} and Nb^{5+} ions results in a substantial reduction in grain size. Consequently, this leads to a decrease in domain size [16]. The smaller the size of the domain, the less resistance it receives when moving, which increases the rate of movement of the domain and enhances the piezoelectric properties of the material. The addition of $\text{Bi}(\text{Zn}_{0.5}\text{Ti}_{0.5})\text{O}_3$ (BZT) may effectively reduce the strain hysteresis in lead-based piezoelectric ceramics, making them suitable for high-precision drives [21]. Moreover, the substitution of the A-site ions in BZT with La^{3+} ions can improve the ferroelectric and piezoelectric properties [22]. In our previous study, the 0.3PT–0.65PZ–0.05BLZT system exhibited a good piezoelectric response. Therefore, the La^{3+} and Nb^{5+} ionic co-doping method was used to explore the effects on the

electrical characteristics of 0.3PT–0.65PZ–0.05BLZT ceramics.

2. Experimental procedure

2.1. Sample preparation

The chemical formula of La^{3+} and Nb^{5+} ionic co-doped ceramics is 0.3PT–0.65PZ–0.05BLZT–*x* wt% LN (PT–PZ–BLZT–*x*LN, *x* = 0, 3, 4, 5, 6, 7), where LN is an additive composing of a certain proportion of La_2O_3 and Nb_2O_5 . The ceramics were fabricated using the traditional method of solid-state synthesis. First, the raw powders of La_2O_3 (99.9 %) and Nb_2O_5 (99.9 %) mixed at a mass ratio of 3:2 were ball-milled in ethanol for 12 hours, after that, the mixture was sieved to prepare the LN additive. The high-purity raw powders of PbO , TiO_2 , ZrO_2 , Bi_2O_3 , and La_2O_3 were weighed based on the 0.3PT–0.65PZ–0.05BLZT formula, the mixed powders were then combined with ethanol and ball-milled for 16 hours. After that, the mixtures were dried in an oven at 80 °C for 1–2 hours and calcined at 820 °C for 2 hours. The synthesized powders were weighted based on the PT–PZ–BLZT–*x*LN formula and then subjected to secondary ball milling, drying, and sieving processes. Afterward, the granulated powders were pressed into 12 mm thick sheets under the granulation of 5 % polyvinyl alcohol. The sheets were sintered at 600 °C for 4 hours to remove the binder, followed by sintering at 1230 °C for 2 hours to synthesize the ceramic samples. Finally, the ceramic surfaces were polished to maintain a uniform thickness of approximately 0.3 mm. Silver electrodes were then applied to a ceramic sheet.

2.2. Characterization

The crystal structure of PT–PZ–BLZT–*x*LN (*x* = 0, 3, 4, 5, 6, 7)

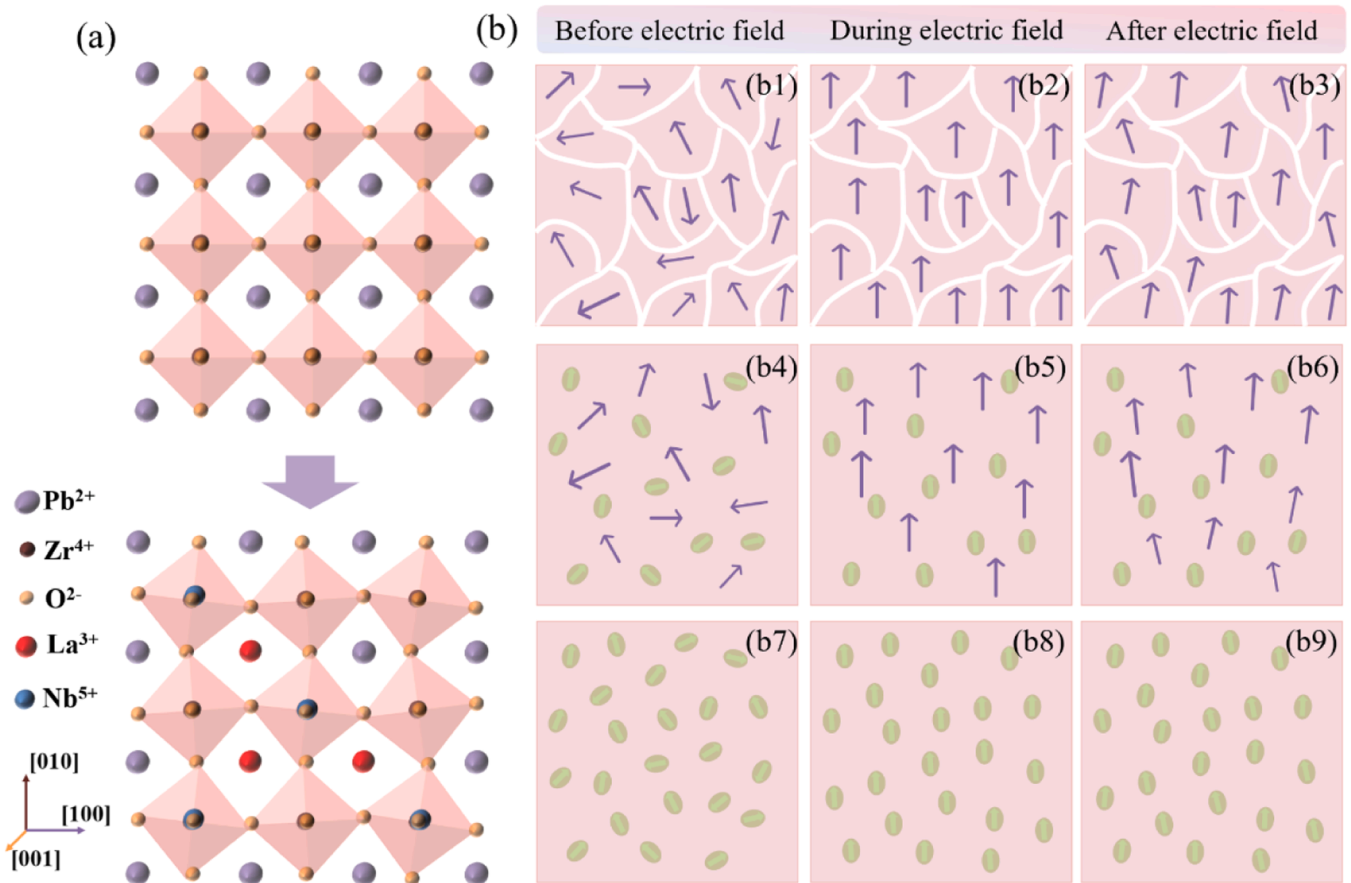


Fig. 1. (a) Lattice distortion diagram of PT–PZ–BLZT–*x*LN ceramics and (b) Behavior of polar nanodomains (PNRs) and ferroelectric macrodomains in the process of applying an electric field.

piezoelectric ceramics was analyzed by X-ray diffractometer (XRD, D8 Advance, Bruker). Rietveld refinement was performed using GSAS II software, with the space group of the structure chosen as $R3m$ [23]. The Raman spectra of the samples in the range of $100\text{--}900\text{ cm}^{-1}$ were tested using a Raman spectrometer (Raman, LabRAM HR, Renishaw, UK). The piezoelectric coefficient of the sample was measured using a piezo- d_{33} m (ZJ-3A, Institute of Acoustics, China) test instrument. The impedance analyzer (Agilent, 4294 A, USA) was used to test the mechanical quality factor Q_m and planar electromechanical coupling coefficient k_p of the samples. A dielectric temperature spectrometer (FZ-IV, Taidao, China) was used to test the dielectric properties of the samples at different frequencies. The hysteresis loop and strain of the samples were measured using a ferroelectric analyzer (4284 A, Agilent, USA) under a frequency of 1 Hz, with the test voltage controlled between 1 kV to 3 kV according to the thickness of the samples. The domain shape at room temperature was observed using piezoresponse force microscopy (PFM, Cypher ES, OXFORD, UK). Scanning electron microscopy (FESEM, SUPRATM55, Germany) was used to observe the microscopic morphology of ceramic samples. The ceramic samples were polished and then chemically etched with 1 % hydrofluoric acid (HF) and hydrochloric acid (HCl). After an initial etching for 1 second, the samples were quickly immersed in a 30 % nitric acid solution (HNO_3) for 3 seconds. After that, the samples were cleaned with deionized water and then dried to observe the ceramic surface morphology.

3. Results and discussion

3.1. Crystal structure and microstructure features

Fig. 2(a) shows the XRD patterns of PT-PZ-BLZT-xLN ($x = 0, 3, 4, 5, 6, 7$) ceramic materials. No other heterophase structures were detected in all samples. Upon examination at a 2θ angle of 44° , the absence of a split in the (200) peak indicates the existence of a rhombohedral (R) phase. A magnification of the (002) diffraction peak, as depicted in Fig. 2(b), reveals that the peak shifts towards a higher angle with an increase in the dopant concentration. This is owing to the different ionic radius of Pb^{2+} ($r = 1.19\text{ \AA}$) and La^{3+} ($r = 1.16\text{ \AA}$) at the A-site, as well as between Zr^{4+} ($r = 0.72\text{ \AA}$), Ti^{4+} ($r = 0.61\text{ \AA}$), and Nb^{5+} ($r = 0.64\text{ \AA}$) at the B-site [24,25]. As a result, cation vacancies are formed within the PT-PZ-BLZT-xLN ceramics to sustain electrical neutrality. These vacancies facilitate the movement of domain walls, thus enhancing the piezoelectric performance.

To further investigate the factors influencing strain, the samples were analyzed by Rietveld refinement analysis using GSAS-II software to analyze the phase composition in detail. Fig. 2(d-i) displays the XRD profiles for the PT-PZ-BLZT-xLN ceramics ($x = 0, 5, 7$) after whole pattern fitting. The results of the test samples can be found in Fig. S1. Through the refinement results, all the samples showed a single rhombohedral phase structure. Table 1 shows the lattice parameters after structural refinement. The lattice parameters and cell volume show some changes, indicating that the crystal structure has experienced minor deformations. Specifically, the oxygen octahedra may distort and cause the lattice to contract after adding the dopants.

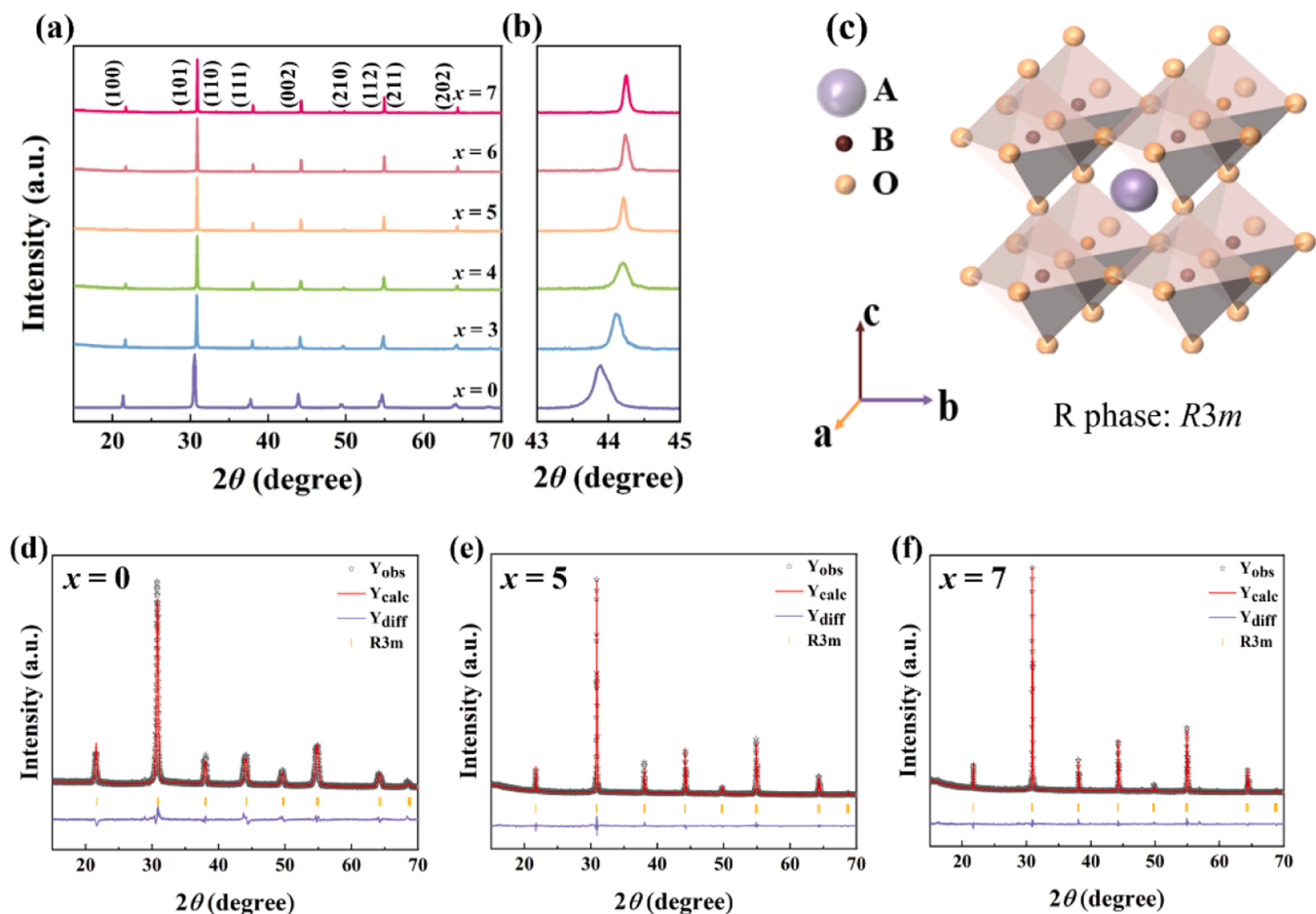


Fig. 2. (a) XRD patterns of the PT-PZ-BLZT-xLN ceramics; (b) Partial enlarged view of the XRD pattern; (c) Crystal structures of rhombohedral (R) phase; (d-f) Rietveld refinement.

Table 1

Structural parameters of PT-PZ-BLZT-xLN ceramics after Rietveld refinement.

Samples	<i>a</i> (Å)	<i>c</i> (Å)	Cell volume (Å ³)	<i>R</i> _{wp}	<i>R</i> _p	χ ²
<i>x</i> = 0	5.8159	14.3043	417.580	5.9 %	4.3 %	2.56
<i>x</i> = 3	5.8129	14.2801	413.884	5.5 %	4.1 %	2.23
<i>x</i> = 4	5.7903	14.2060	412.491	5.2 %	3.9 %	2.21
<i>x</i> = 5	5.7875	14.1797	411.329	5.5 %	4.1 %	2.27
<i>x</i> = 6	5.7845	14.1667	410.519	5.8 %	4.2 %	2.41
<i>x</i> = 7	5.7840	14.1647	410.411	5.2 %	3.9 %	2.21

Furthermore, the structural distortion of piezoelectric ceramics can be reflected by the Goldschmidt tolerance factor (*t*). The *t* value is calculated by Eq. (1) [26–28], which can be explained by the general formula ABO₃.

$$t = \frac{R_A + R_B}{\sqrt{2}(R_B + R_O)} \quad (1)$$

where *R*_A, *R*_B, and *R*_O represent the ionic radii of Pb²⁺ and La³⁺, the ionic radius of Nb⁵⁺, Ti⁴⁺, and Zr⁴⁺, and the radius of O²⁻, respectively. According to Eq. (1), it is evident that the substitution of La³⁺ (with an ionic radius of 1.16 Å) for Pb²⁺ (with an ionic radius of 1.19 Å) results in a decrease in the molecular dimensions as described in Eq. (1), consequently leading to a reduction in the *t* value. This reduction in the tolerance factor decreases the stability of the perovskite structure and results in lattice distortion. However, a lower tolerance factor can lead to a more distorted perovskite structure, which may enhance certain properties, such as piezoelectricity, by making the domain wall easier to move [29].

The Gaussian and Lorentz fitting methods were applied to the Raman spectral data of PT-PZ-BLZT-xLN by using the Peakfit software, as

shown in Fig. 3(a). Also, the wavenumber fluctuation about certain Raman modes is depicted in Fig. 3(b). The Raman mode below 200 cm⁻¹ is related to the vibrational motion occurring at the A-site within the perovskite crystal lattice [30]. The peak at 125 cm⁻¹ undergoes a red shift as *x* increases. This phenomenon may be attributed to the cationic migration along the crystallographic axis, which could be linked to the displacement of La³⁺ ions at the non-centrosymmetric A-site. The vibrational behavior of the B–O bonds and the [BO₆] octahedra are manifested in the Raman modes within the wavenumber ranges of 200 cm⁻¹ to 400 cm⁻¹ and 400 cm⁻¹ to 700 cm⁻¹, respectively. The distortions of these B–O bonds and [BO₆] octahedra are closely associated with the dynamical properties of the polar nanoregions (PNRs) within the perovskite framework, which reflects the relaxation contributions from the B-site. As illustrated in Fig. 3(c), the increment in the doping concentration of *x* results in a gradual decrease in the B–O bond length. The bond strength of the B–O bond is quantified through the Eq. (2).

$$S = \left(\frac{R}{R_1}\right)^{-N} \quad (2)$$

where *R* denotes the refined bond length of the B–O bond, *R*₁ = 1.907, and *N* = 5 represent the standard bond length and bond length parameters, respectively [31]. The computational results indicate an enhancement in the bond strength between the B–O bonds, suggesting that the contraction of the [BO₆] octahedra and the shortening of the B–O bond are the result of the incorporation of Nb⁵⁺ at the B-site. Consequently, this led to a strengthening of the B–O bond. This strengthening induces a blue shift within the 200–400 cm⁻¹ range. The enhanced bond strength of the B–O bonds contributes to the stabilization of the relaxation state and the non-polar state within the system.

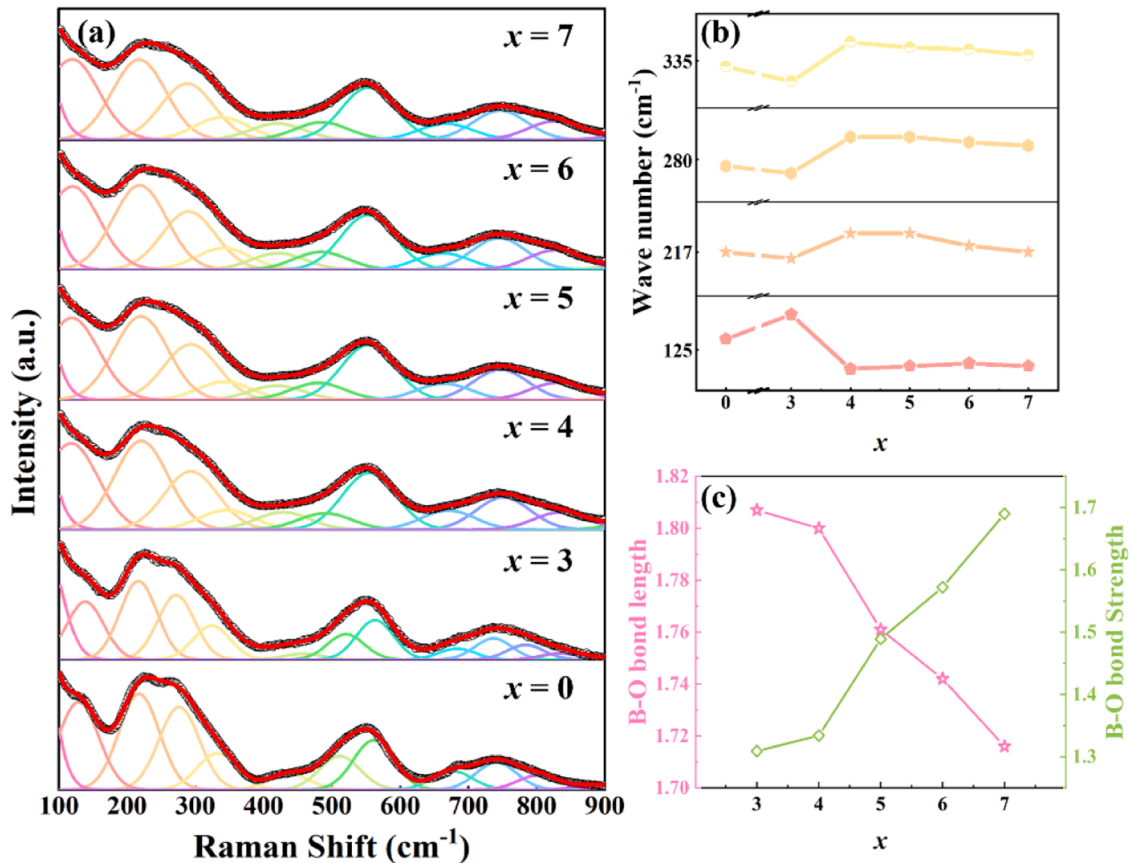


Fig. 3. (a) The Raman peaks of the PT-PZ-BLZT-xLN ceramic at room temperature; (b) The frequency shift of the corresponding modal peaks. (c) Refined B–O bond length and calculated B–O bond strength for PT-PZ-BLZT-xLN ceramics.

Fig. 4(a-f) displays the SEM images of the ceramics sintered at 1230 °C. All the samples have full grains with no significant impurities (the cross-section view of Fig. 4(a) is detailed in Fig. S2). Measuring the average grain size (AGS) of these samples using the line intercept method using nanoscale measurement software allows for an in-depth study of the effect of grain size on piezoelectric properties [32]. The AGS values are 10.63 μm , 1.63 μm , 2.00 μm , 2.38 μm , 2.50 μm , and 2.71 μm when the doping levels of x reach 0, 3, 4, 5, 6, and 7, respectively. The addition of dopants leads to a significant reduction in grain size [31]. The reduction of the 90° domain within is facilitated by the small grain size [33]. Consequently, the activity of these 90° domains increases. At this point, an electric field is applied, and the electrical domains can be flipped at a smaller amount of energy, increasing the number of domain reversals [29]. This phenomenon is reflected as an enhancement in the piezoelectric coefficient and an improvement in the material's piezoelectric properties [34]. Fig. 5 depicts the Energy-Dispersive Spectroscopy (EDS) spectrum of the sample at $x = 5$. The spectrum reveals a homogeneous distribution of elements, such as Pb, Zr, Ti, Nb, La, Bi, and Zn throughout the ceramic grains.

Nb has a higher electronegativity (1.6) than Ti (1.54) and Zr (1.33), hence, the B–O bond in [BO₆] octahedron may be shortened by the strong interaction between Nb and O. This disruption of the macroscopic ferroelectric domains can promote the appearance of microdomains [35]. When the size of microdomains decreases to nanoscale, PNRs can be found. The aggregation of numerous microdomains results in the formation of local heterogeneous structures [36]. These microdomains are more responsive to external electric fields due to their low internal

stress. To directly observe the effect of La^{3+} and Nb^{5+} co-doping on domain size, PFM is utilized to study the domains in ceramic samples at $x = 0$ and $x = 5$, with the results shown in Fig. 6.

In the domain at $x = 0$, the structure is characterized by the presence of small, uniformly distributed parallel fringe domains. Conversely, the piezoelectric ceramic at $x = 5$ displays a complex, labyrinthine surface domain pattern with domains oriented randomly. Numerous studies have demonstrated that smaller domain sizes or higher domain wall densities are essential for enhanced piezoelectric performance, as referenced in [37] and [38]. Consequently, the findings of this study follow the fundamental principles of domain dynamics, as elucidated in Ref. [39]. The labyrinthine domain structure with reduced dimensions is instrumental in conferring the superior piezoelectric properties observed in PT–PZ–BLZT–xLN piezoelectric ceramics, as further detailed in Ref. [40].

3.2. Dielectric properties analysis

Fig. 7(a) depicts the dielectric properties of the PT–PZ–BLZT–xLN ceramic samples. The results indicate that there is a steady decline in the relative dielectric constant, designated as ε_r , of the samples. Eq. (3) is used to further analyze how the co-doping of La^{3+} and Nb^{5+} affects the relaxor behavior of ceramics [41,42].

$$\frac{1}{\varepsilon} - \frac{1}{\varepsilon_m} = \frac{(T - T_m)^\gamma}{C} \quad (3)$$

In the equation, ε_{max} is the maximum relative permittivity, γ represents the dispersion factor, the dispersion factor is denoted by C , which is a constant, and T_{max} is the temperature that the piezoelectric ceramics exhibit the maximum relative permittivity. For a normal ferroelectric,

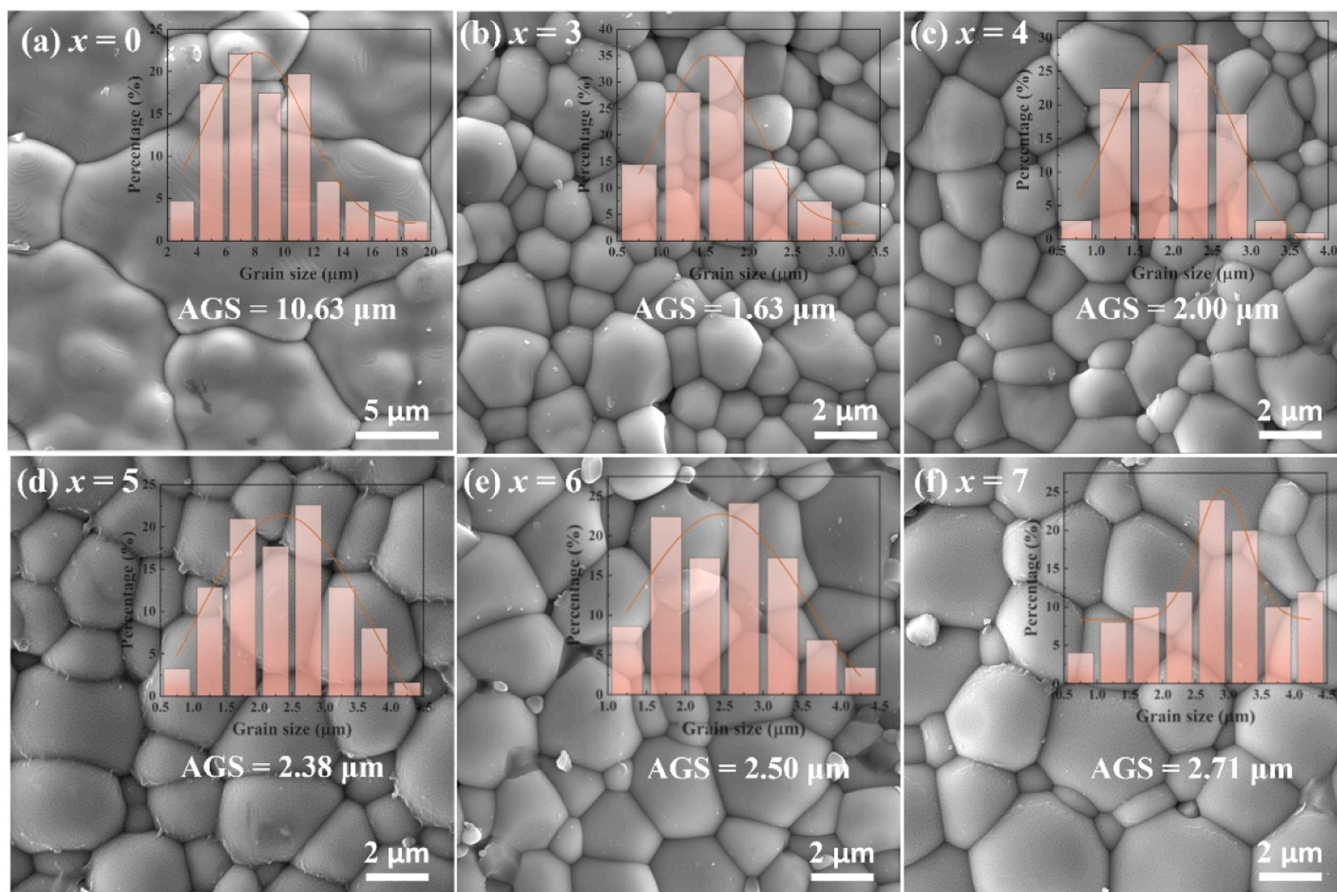


Fig. 4. . (a-f) SEM images of the PT–PZ–BLZT–xLN samples: (a) $x = 0$, (b) $x = 3$, (c) $x = 4$, (d) $x = 5$, (e) $x = 6$, (f) $x = 7$.

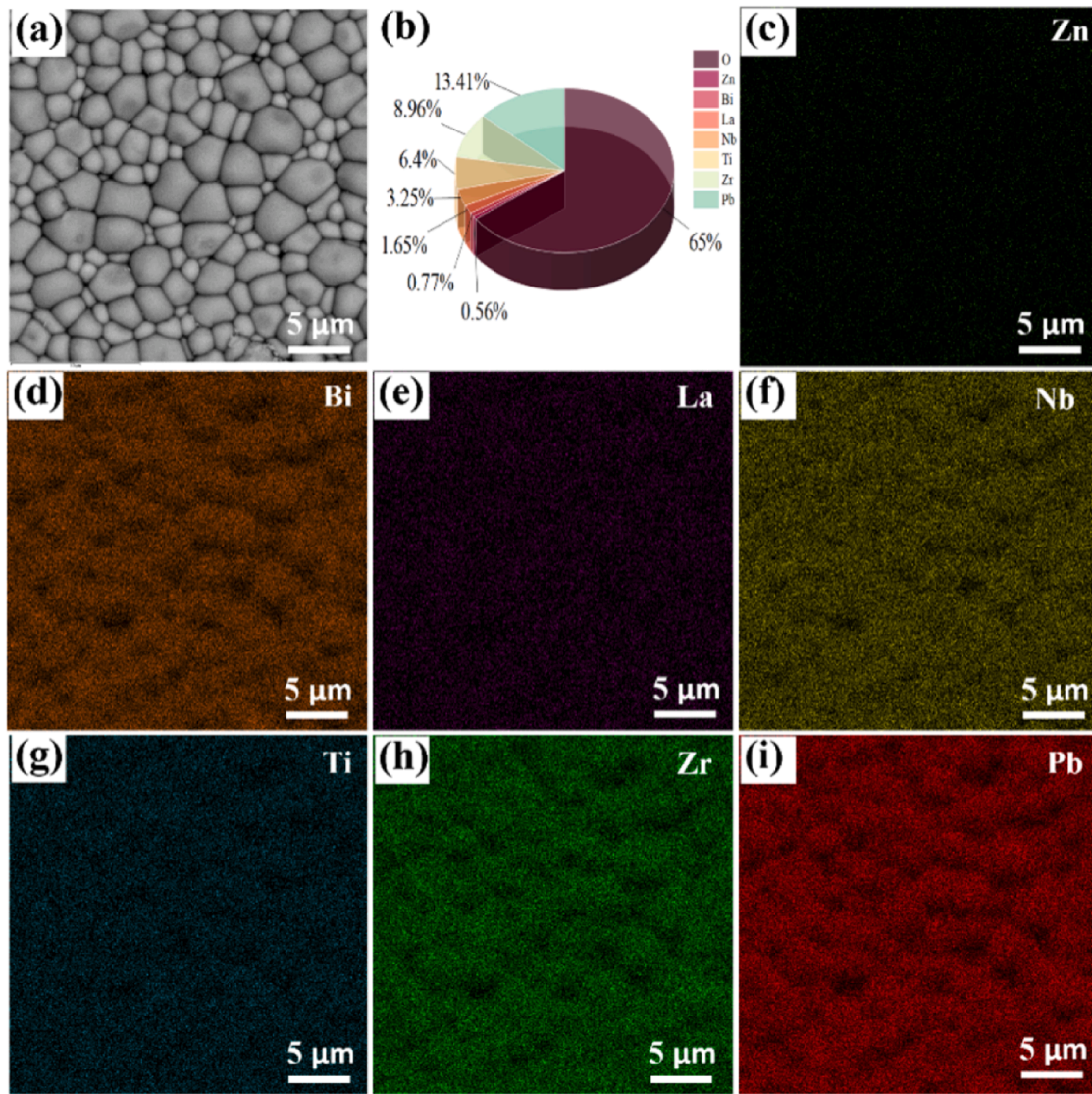


Fig. 5. (a) The original surface of the sample, (b) Proportion of different atoms, (c-i) EDS mapping results of PT-PZ-BLZT-5LN ceramics.

the γ value ranging from 1 to 2 determines the degree of ferroelectric relaxation. The larger the γ , the stronger the relaxation ferroelectric behavior. When the doping amount increases to $x = 5$, the γ value increases from 1.30 to 1.82, which indicates a substantial enhancement in the relaxor properties of the PT-PZ-BLZT- x LN piezoelectric ceramics. The doping of La^{3+} and Nb^{5+} may have disrupted the original electrical domains, leading to the emergence of microdomains. Additionally, the introduction of La^{3+} and Nb^{5+} may strengthen the disorder of cations, forming smaller PNRs. As a result, the relaxation behavior of piezoelectric ceramics is enhanced [43–45].

It has been reported that when La^{3+} enters the lattice, the introduction of La^{3+} replaces Pb^{2+} at the A-site and creates vacancies, as shown in Eq. (4):



The defect chemistry resulting from the doping of high-valent dopants at the A-site significantly impacts the relaxation properties of piezoelectric ceramics [46–51]. The T_c , ϵ_r , and $\tan\delta$ values of the PT-PZ-BLZT- x LN ceramics are shown in Fig. 7(c). As the doping amount of La^{3+} and Nb^{5+} ions increase, both the Curie temperature and the relative permittivity reduce. An increase in doping content tends to lower the T_c . Although the T_c is reduced, it remains at a normal level. It is

noteworthy that the minimum dielectric loss is achieved when $x = 5$.

As seen in Fig. 7(d-i), the observed samples demonstrate typical relaxor ferroelectric characteristics, meaning that the change in the ϵ_r near the dielectric maximum value T_m shows a broad curve. This may be related to nanoscale heterogeneity caused by cationic disorder [52,53]. The findings suggest that the dielectric peaks of the samples all shift towards the right, and the peaks decrease with the increase in frequency. This demonstrates a significant frequency dispersion phenomenon. These findings further prove the relaxor behavior of the ceramics.

3.3. Ferroelectric properties analysis

Fig. 8(a-f) illustrates the trend of P - E loops and J - E curves of ceramic samples. When the doping content level is small, the P - E loops exhibit relatively saturated curves. This is because co-doping with La^{3+} and Nb^{5+} ions likely alters the balance of energy states within the material, consequently affecting the characteristics. Additionally, the coercive field (E_c) is reduced from $8.85 \text{ kV}\cdot\text{cm}^{-1}$ to $4.86 \text{ kV}\cdot\text{cm}^{-1}$, and the remanent polarization (P_r) decreases from $26 \mu\text{C}/\text{cm}^2$ to $6 \mu\text{C}/\text{cm}^2$. The reduction in P_r and E_c is attributed to the change in the content of the tetragonal, which leads to a reduction in lattice deformation resistance and facilitates domain wall motion. The maximum polarization (P_{max}) also decreases from $35.06 \mu\text{C}/\text{cm}^2$ to $30 \mu\text{C}/\text{cm}^2$. The analysis of

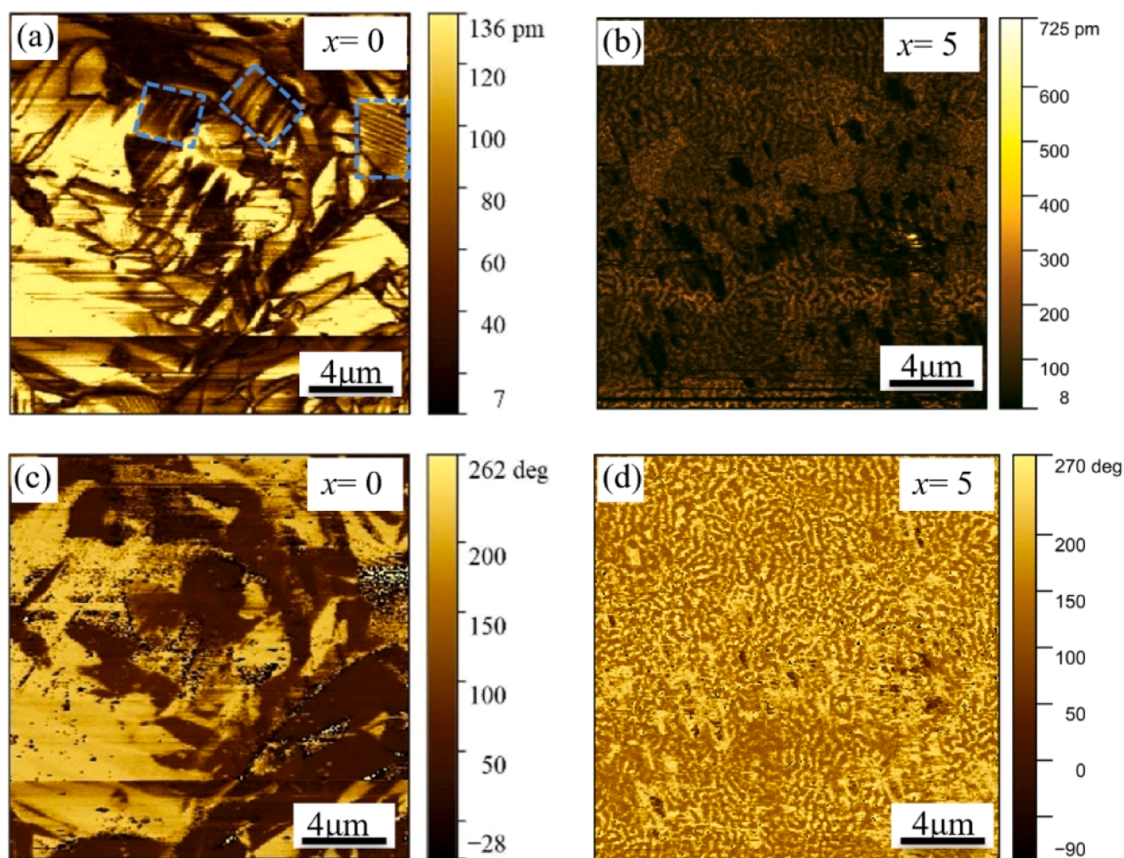


Fig. 6. PFM amplitude and phase images of the PT-PZ-BLZT-xLN ceramics: (a, c) $x = 0$, (b, d) $x = 5$.

dielectric behavior reveals a decrease in P_r and E_c , coupled with enhanced dielectric relaxor characteristics, suggesting a transition in the morphology of electrical domains.

The J - E curve for the undoped ceramic sample is characterized by a slender, single peak, which is predominantly attributed to the contribution of ferroelectric macrodomains. With an increase in the doping level, this single peak progressively broadens and decreases in height. At $x = 5$, the current peak splits into two distinct peaks, as depicted in Fig. 8 (d). Typically, in ferroelectric materials, the split of the current peak is associated with defective dipoles arising from the domain structure, polarization reversal behavior, and non-equivalent substitutions within the material. These factors can lead to the emergence of multiple current peaks, contributing to the strain, as reported in Refs. [54–57]. As illustrated in Fig. 6, the transformation of the electrical domain from a large ferroelectric macrodomain to a smaller labyrinth domain may lead to the splitting of current peaks. This indicates a high degree of activity and rapid response to an external field, which significantly enhances the material strain, as discussed in Ref. [55]. Moreover, non-equivalent substitutions, such as Pb^{2+} vacancies, can result in charge enrichment and the formation of defective dipoles, as described in Refs. [56,57]. These defective dipoles can drive the formation of a charged polarized topological center, influencing the polarization reversal behavior of the material and leading to the emergence of peak splitting.

Fig. 9(a–f) displays the strain of the PT-PZ-BLZT-xLN ($x = 3, 5, 7$) piezoelectric ceramics. The graphs of the rest samples are presented in Fig. S3. The reduction in domain size is beneficial to enhance the strain value [18]. It can be observed that the unipolar strain increases from 0.3 % to 0.45 % after the co-doping of La^{3+} and Nb^{5+} ions.

Table 2 presents a comparative analysis of the performance of piezoelectric ceramics as reported in different studies, where the rest of the properties can be found in Fig. S4. The piezoelectric ceramics synthesized in this study demonstrate great piezoelectric properties,

including strain, strain hysteresis, and the d_{33} piezoelectric coefficient. These findings indicate the potential of the developed ceramics applied in the field of piezoelectric actuation.

4. Conclusions

This study conducted a detailed investigation on the influences of La^{3+} and Nb^{5+} ions co-doping on the relationship between the structure and properties of PT-PZ-BLZT piezoelectric ceramics. The results indicate that the co-doping of La^{3+} and Nb^{5+} ions has induced significant lattice distortion that is reflected by the variations of structural parameters. This changes the internal stress, making domains easier to move. The co-doping of La^{3+} and Nb^{5+} ions decreases the average grain size, which may result in a decrease in domain size. Through the PFM analysis, the co-doping strategy shortens the B–O bond length and then forms the microdomains. This is favorable for the generation of large strains. A relaxor behavior observed from the dielectric temperature spectrum is mainly caused by the formation of microdomains. Meanwhile, the La^{3+} ion can substitute for the Pb site and form Pb^{2+} vacancies, which also results in the relaxor characteristics and improves the d_{33} value (585 pC/N). Benefitting from the formation of microdomains, the La^{3+} and Nb^{5+} ionic co-doped PT-PZ-BLZT-based ceramics achieve a high strain (0.45 %) and low hysteresis (16 %). These substantial field-induced strains and minimal hysteresis render this material a viable candidate for high-strain actuator applications.

CRediT authorship contribution statement

Yumei Tan: Writing – review & editing, Writing – original draft, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Yinghao Zhu:** Software, Methodology, Investigation. **Hongyu Yang:** Writing – review & editing, Supervision, Resources, Project

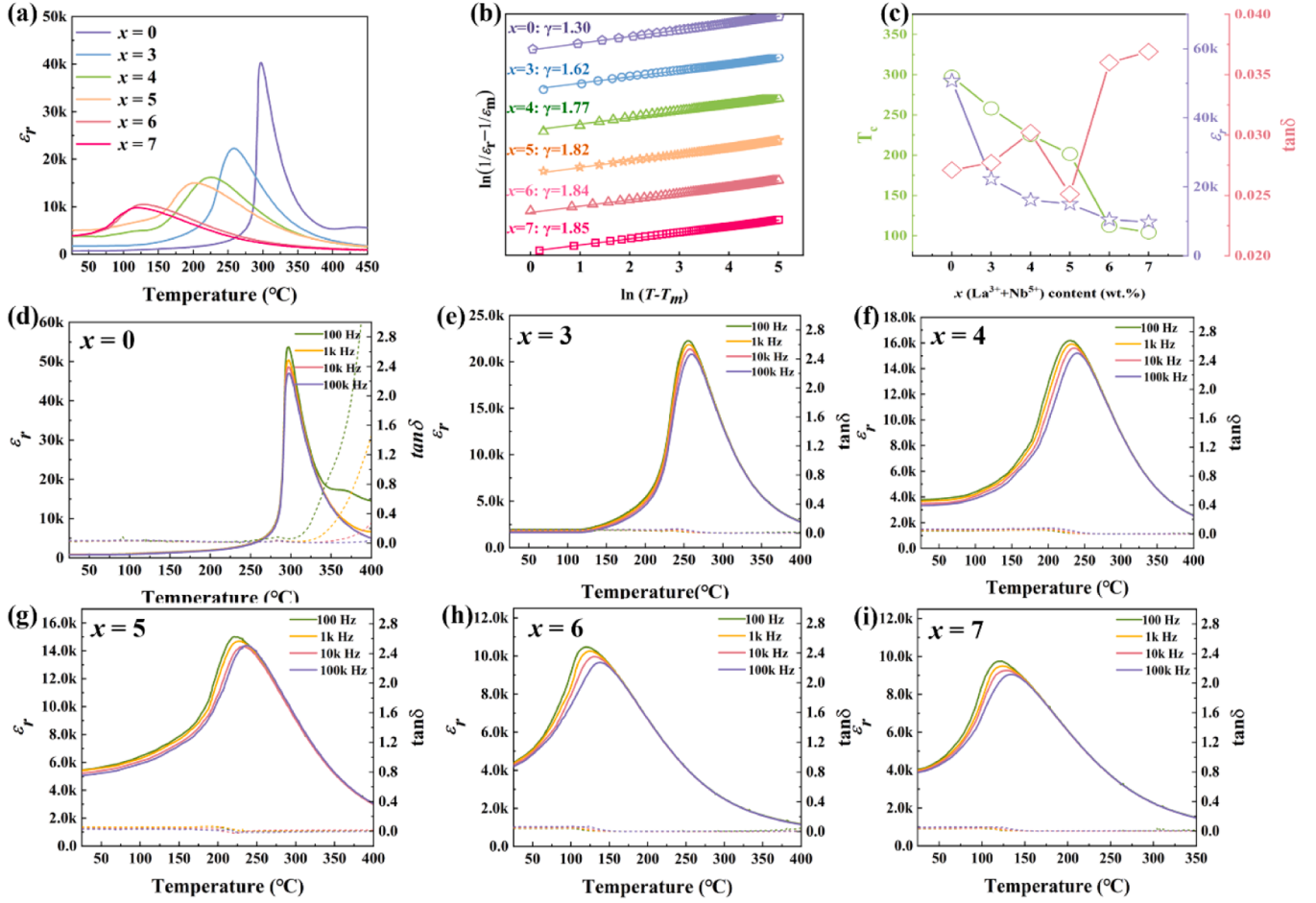


Fig. 7. (a) Dielectric temperature spectrum of PT-PZ-BLZT-xLN piezoelectric ceramics at 1 kHz; (b) $\ln(1/\epsilon_r - 1/\epsilon_{\max})$ vs. $\ln(T - T_m)$ curves; (c) Variations of T_c , ϵ_r , and $\tan\delta$ value. (d-i) Dielectric temperature spectra of the PT-PZ-BLZT-xLN ceramics under different conditions.

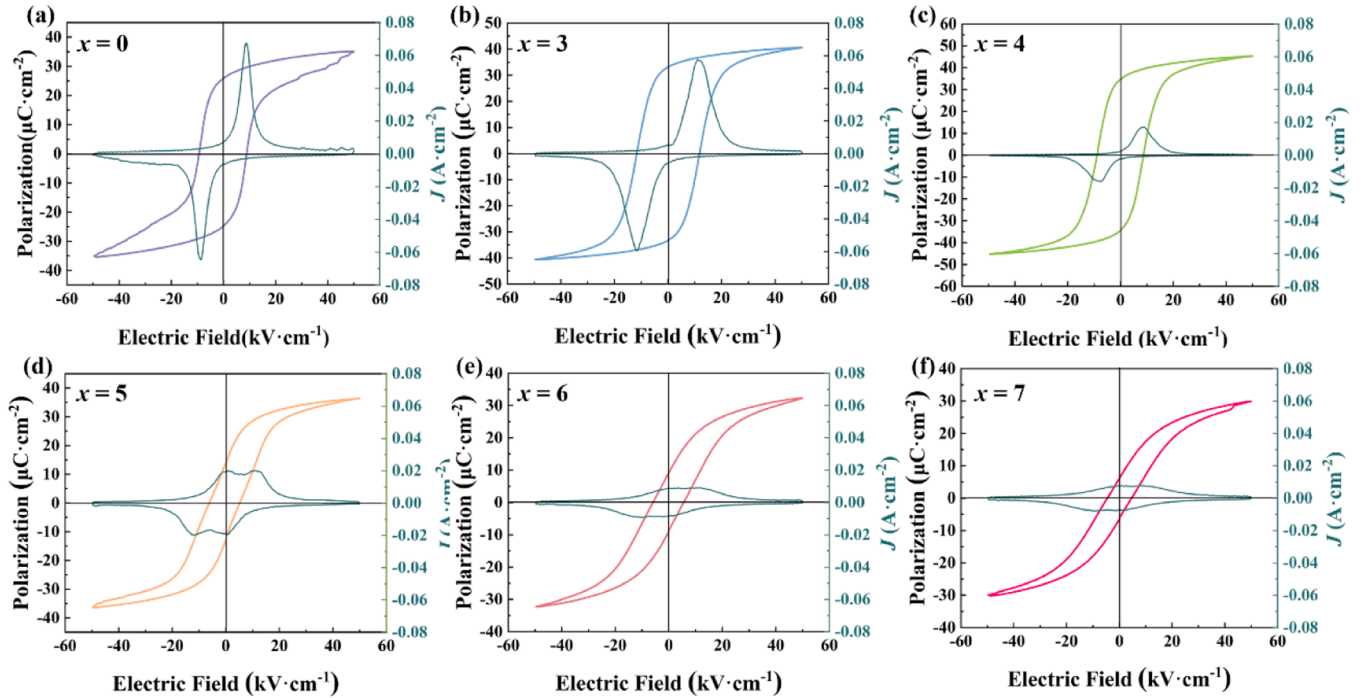


Fig. 8. The electric P - E loops and J - E curves of the PT-PZ-BLZT-xLN piezoelectric ceramics: (a) $x = 0$, (b) $x = 3$, (c) $x = 4$, (d) $x = 5$, (e) $x = 6$, (f) $x = 7$.

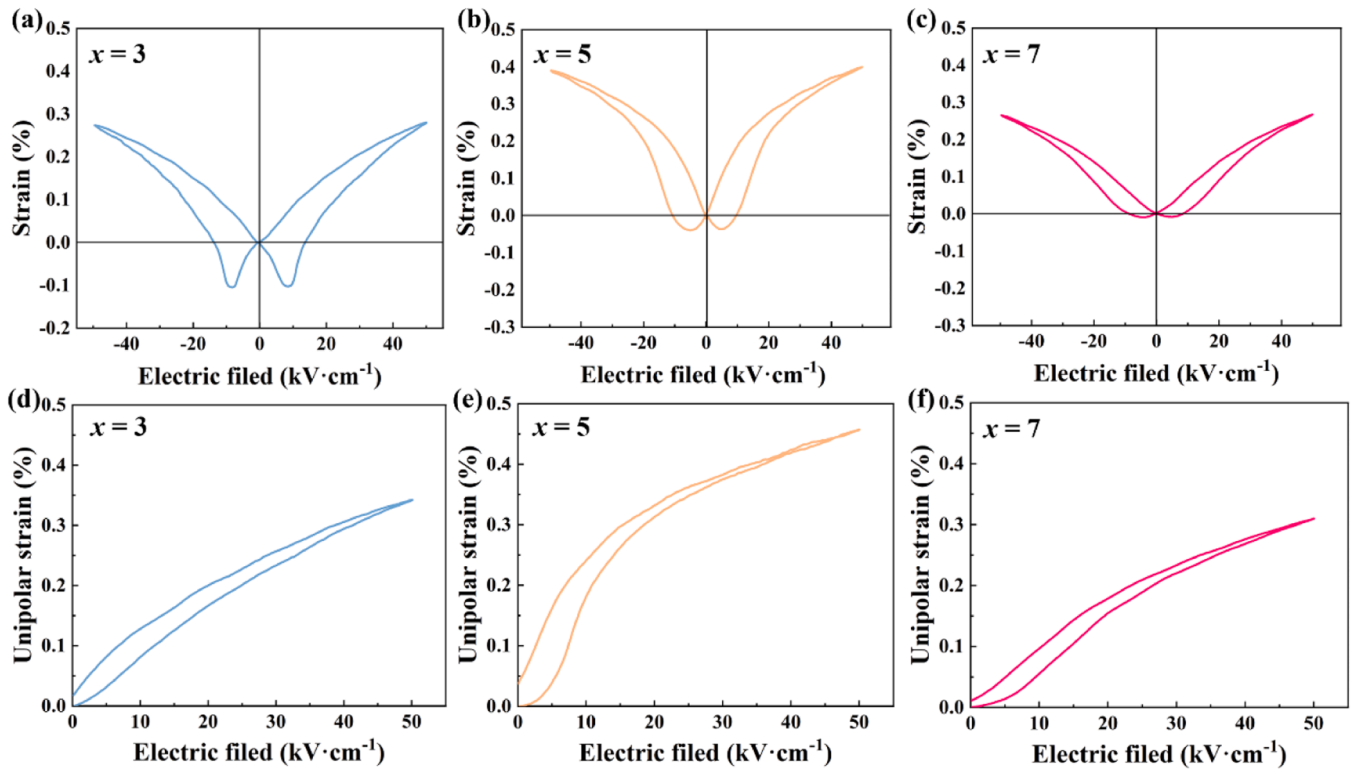


Fig. 9. The butterfly curve and unipolar strain graph of the PT-PZ-BLZT-xLN ceramics: (a, d) $x = 3$, (b, e) $x = 5$, (c, f) $x = 7$.

Table 2
Performance comparison of piezoelectric ceramics.

Materials	E (kV/cm)	Strain	Hysteresis	d_{33} (pC/N)	Refs.
PNN-PZT-Sn	50	0.06 %	28 %	1123	[58]
PNN-PZT-Ta	15	0.12 %	25 %	1090	[59]
BZT-PZ-PT	40	0.13 %	40 %	341	[21]
PHT-PIN-Li/W	30	0.15 %	30 %	655	[28]
BS-PT-Mn	40	0.18 %	32 %	-	[60]
BS-PT-PMN-Fe	40	0.23 %	36 %	160	[61]
BS-PT	40	0.25 %	50 %	-	[60]
BMT-PMN-PT	60	0.29 %	18 %	120	[60]
PT-PZ-BNT	50	0.40 %	38 %	200	[62]
Textured PMN-24 % PT	30	0.10 %	0 %	-	[63]
PT-PZ-BLZT-5LN	50	0.45 %	16 %	585	This work

administration, Funding acquisition, Data curation. **Zhimin Li**: Writing – review & editing, Supervision, Resources, Project administration, Funding acquisition, Data curation. **Denghui Jiang**: Methodology, Formal analysis, Data curation. **Kao Pei**: Software, Investigation, Data curation. **Zhaoru Xue**: Visualization, Validation, Software, Methodology, Investigation. **Linzhuang Xing**: Writing – review & editing, Resources, Project administration.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at [doi:10.1016/j.jallcom.2024.177560](https://doi.org/10.1016/j.jallcom.2024.177560).

Data Availability

Data will be made available on request.

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